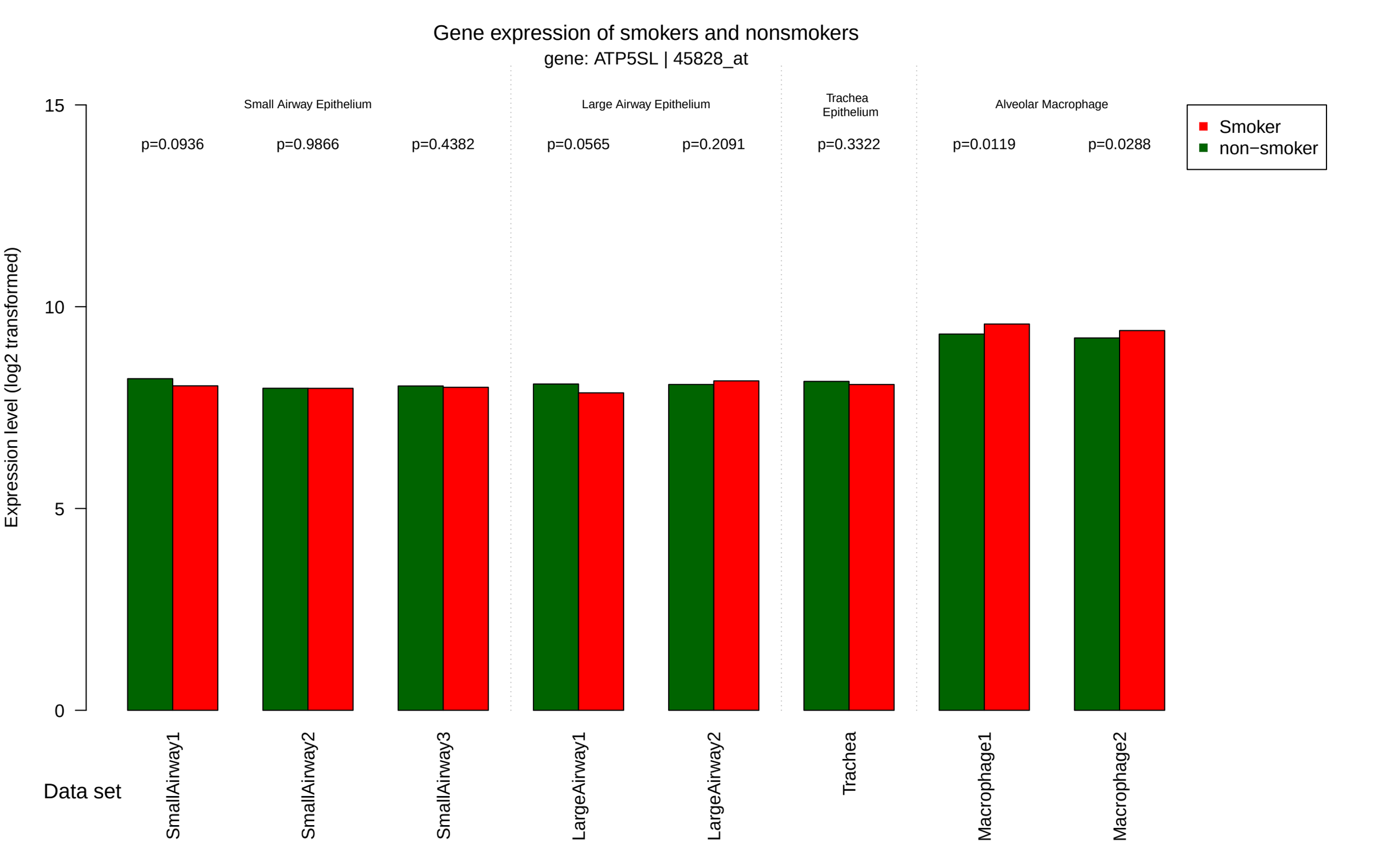
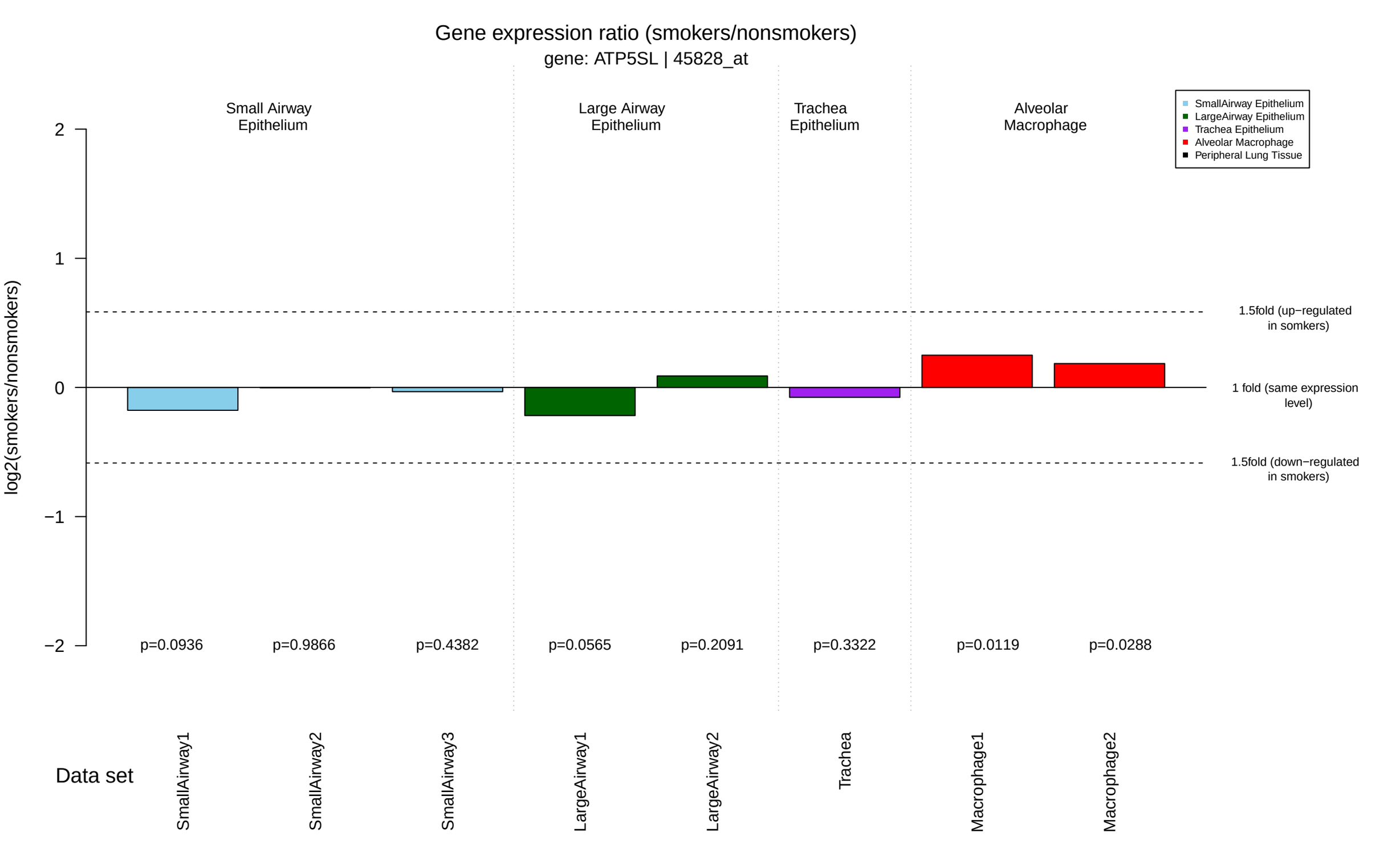
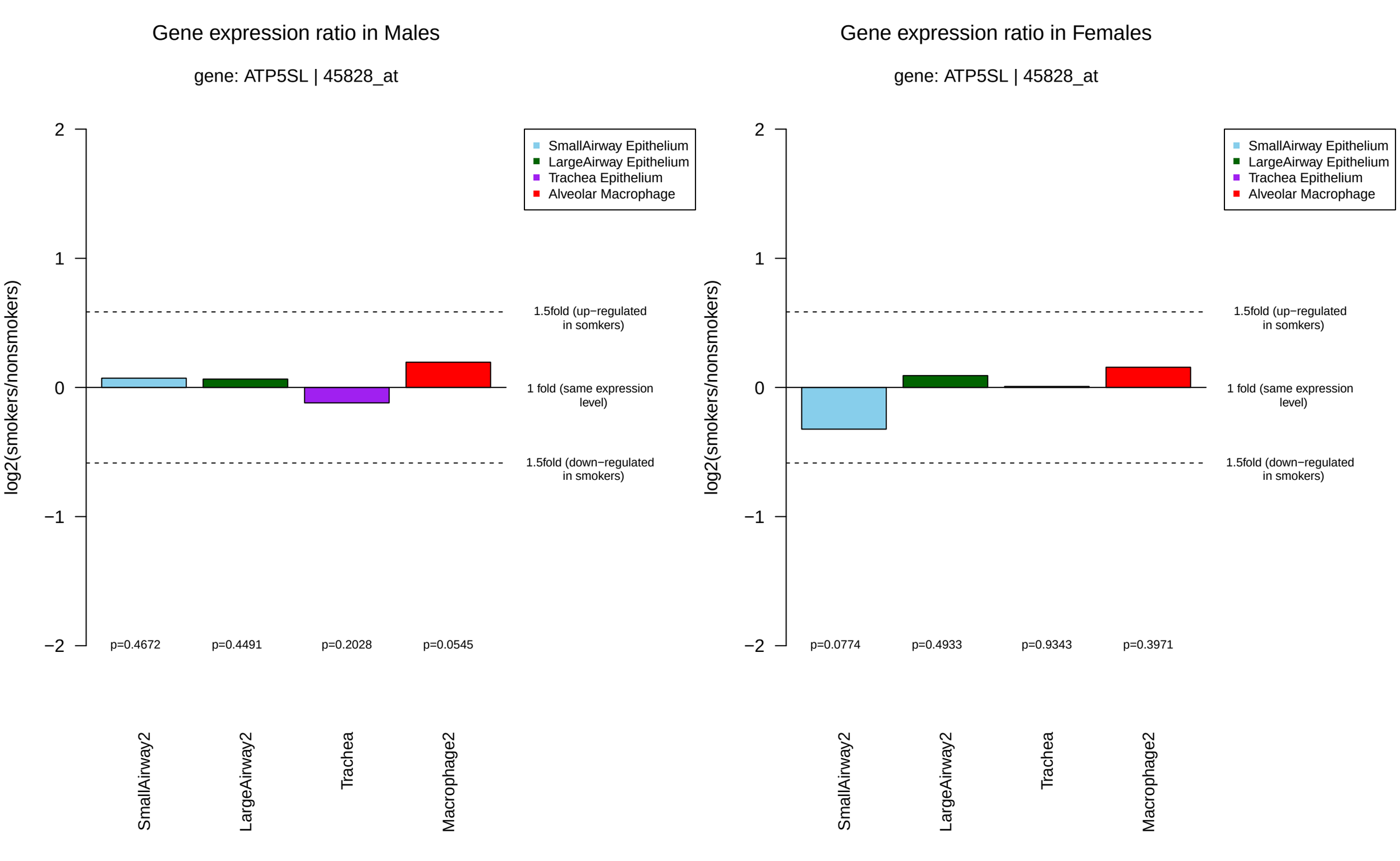
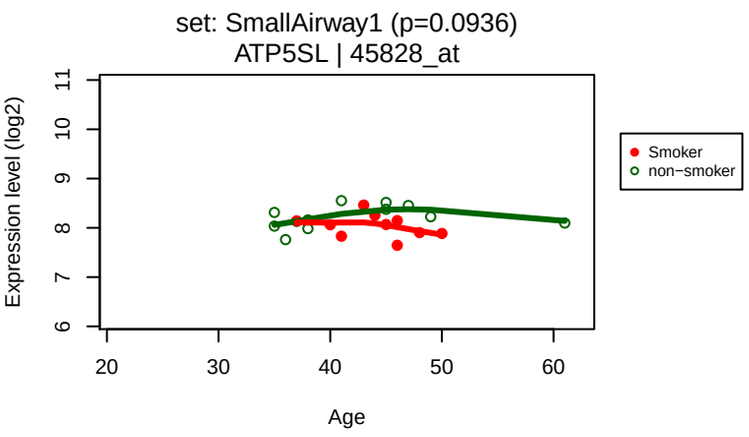
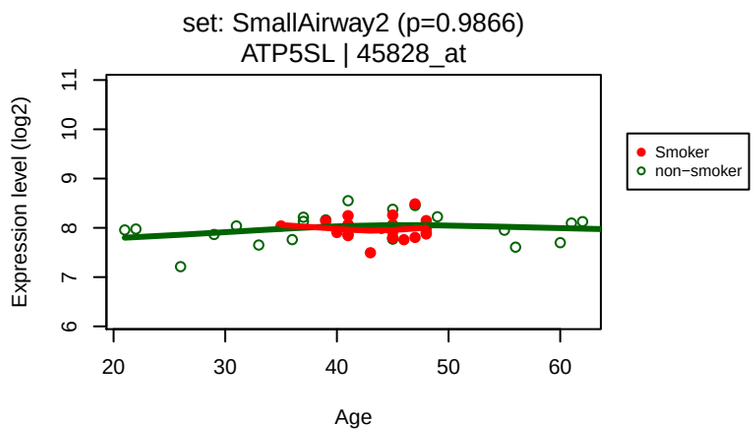
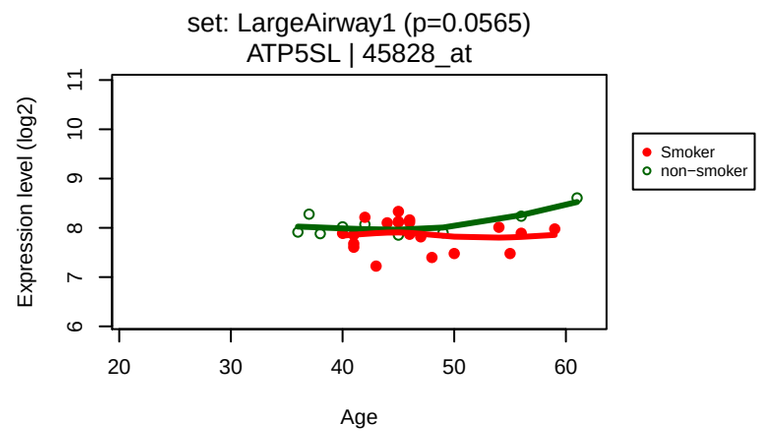

Gene expression of smokers and nonsmokers
gene: ATP5SL | 45828_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0936
p=0.9866
p=0.4382
p=0.0565
p=0.2091
p=0.3322
p=0.0119
p=0.0288
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

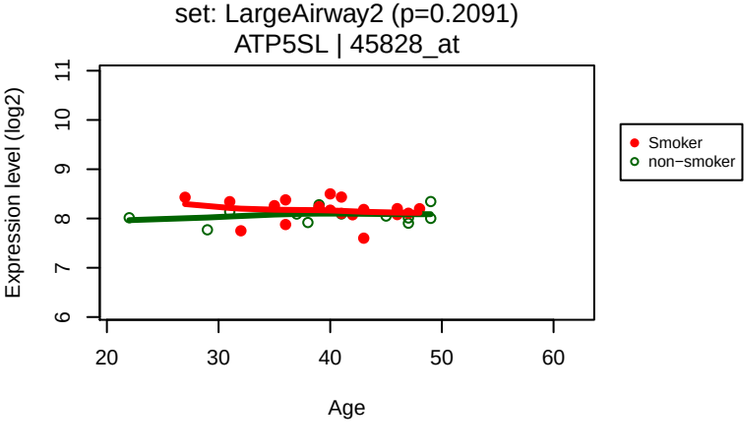

Gene expression ratio (smokers/nonsmokers)
gene: ATP5SL | 45828_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0936
p=0.9866
p=0.4382
p=0.0565
p=0.2091
p=0.3322
p=0.0119
p=0.0288
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

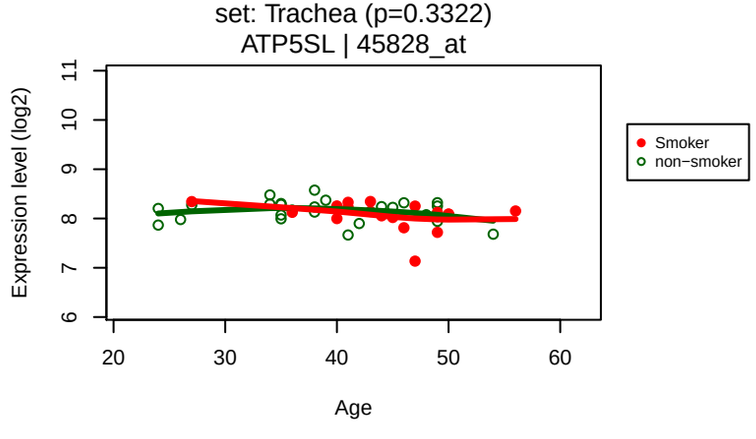

Gene expression ratio in Males
gene: ATP5SL | 45828_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4672
p=0.4491
p=0.2028
p=0.0545
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP5SL | 45828_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0774
p=0.4933
p=0.9343
p=0.3971
SmallAirway2
LargeAirway2
Trachea
Macrophage2

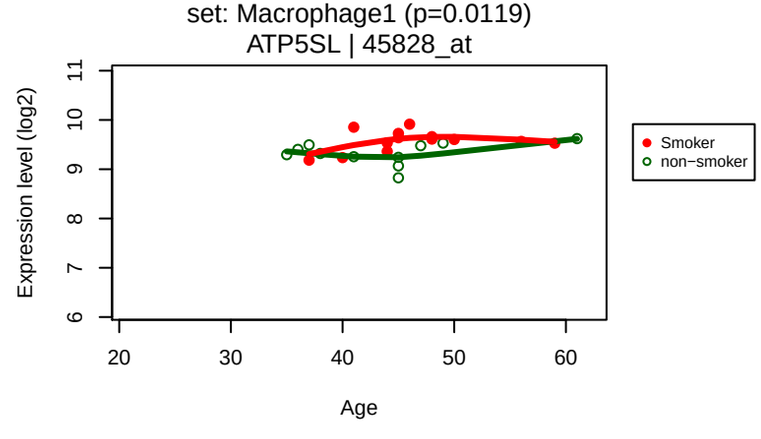

set: SmallAirway1 (p=0.0936)
ATP5SL | 45828_at
Expression level (log2)
Age
Smoker
non−smoker

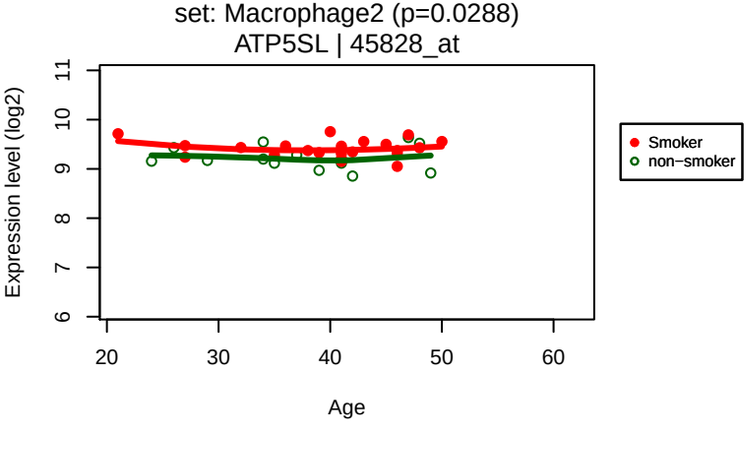

set: SmallAirway2 (p=0.9866)
ATP5SL | 45828_at
Expression level (log2)
Age
Smoker
non−smoker

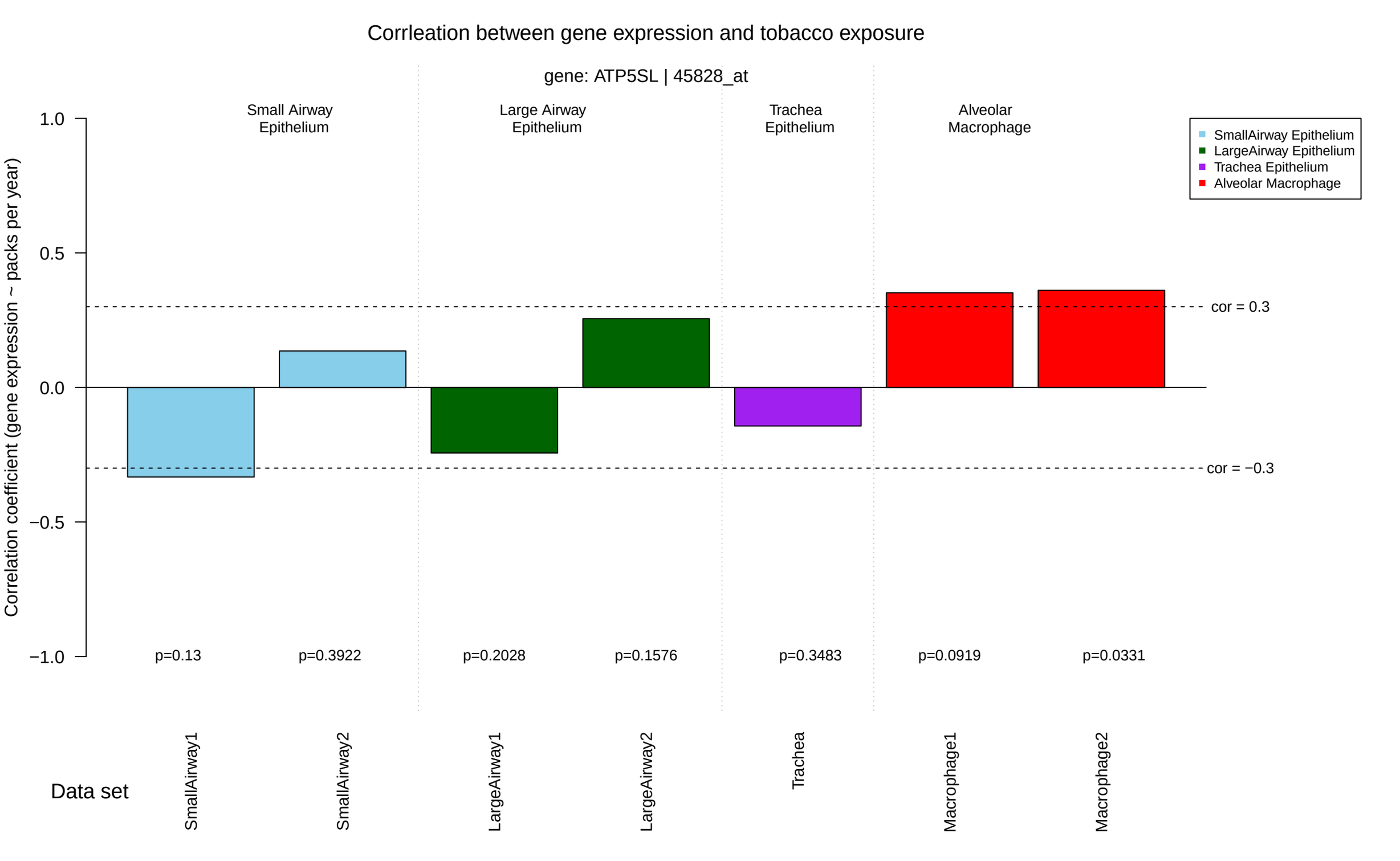

set: LargeAirway1 (p=0.0565)
ATP5SL | 45828_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2091)
ATP5SL | 45828_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3322)
ATP5SL | 45828_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0119)
ATP5SL | 45828_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0288)
ATP5SL | 45828_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP5SL | 45828_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.13
p=0.3922
p=0.2028
p=0.1576
p=0.3483
p=0.0919
p=0.0331
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2